

3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Experiment 3.1:

Addition and Multiplication of Two Sequences

Program:

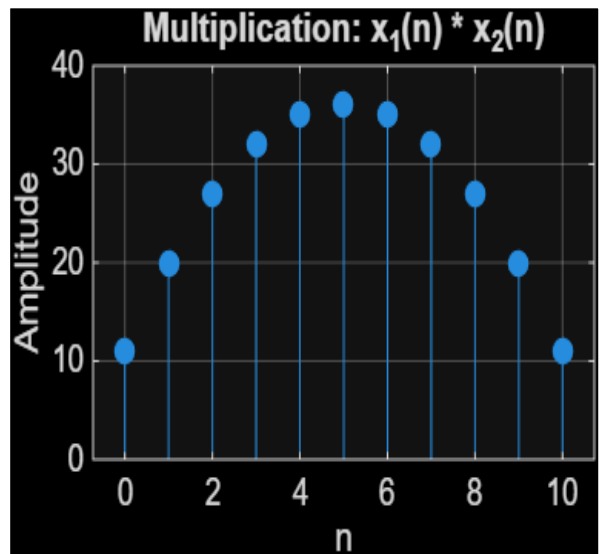
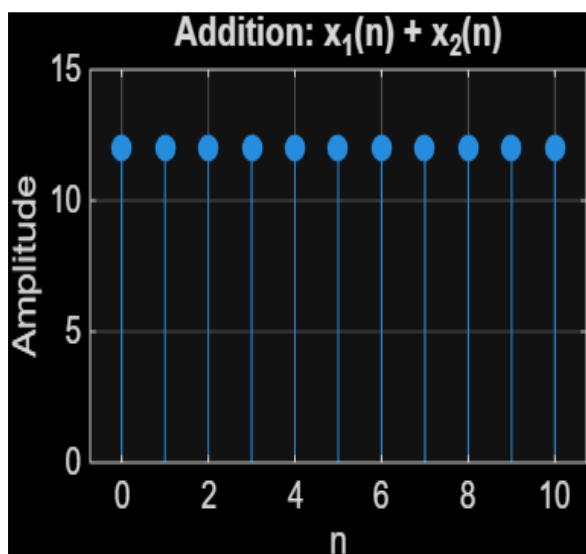
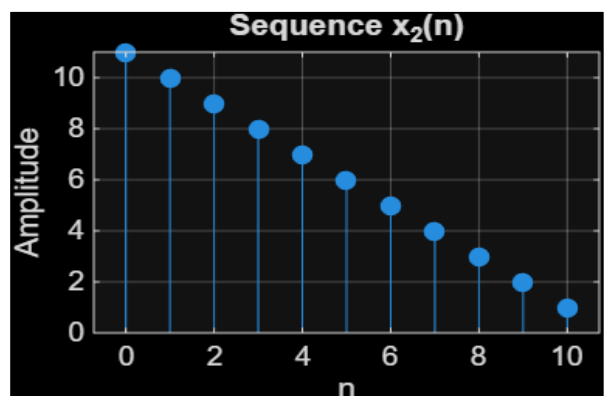
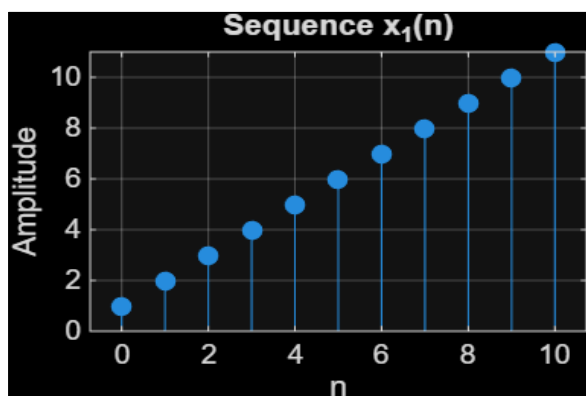
```
clc;
clear;
close all;
% Sample index
n = 0:10;
% Sequence 1
x1 = [1 2 3 4 5 6 7 8 9 10 11];
% Sequence 2
x2 = [11 10 9 8 7 6 5 4 3 2 1];
% Addition of sequences
y_add = x1 + x2;
% Multiplication of sequences
y_mul = x1 .* x2;
% Plotting
figure;
subplot(2,2,1);
stem(n,x1,'filled');
grid on;
title('Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,2);
stem(n,x2,'filled');
grid on;
title('Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,3);
```

```

stem(n,y_add,'filled');
grid on;
title('Addition:  $x_1(n) + x_2(n)$ ');
xlabel('n');
ylabel('Amplitude');
subplot(2,2,4);
stem(n,y_mul,'filled');
grid on;
title('Multiplication:  $x_1(n) * x_2(n)$ ');
xlabel('n');
ylabel('Amplitude');

```

Output Waveform:



Result:

Two discrete-time sequences were generated and their addition and point-wise multiplication were performed successfully using MATLAB. The resulting sequences were displayed using stem plots, illustrating basic operations on discrete-time signals.

Experiment 3.2 :

Energy and Power Calculation of Given Signals

Pogram:

% Energy and Power Calculation of Signals

clc;

clear;

close all;

% Signal

x = [zeros(1,6) 1 2 3 4 5 4 3 2 1 zeros(1,6)];

% Time index

n = -10:10;

% Energy Calculation

E = sum(abs(x).^2);

% Power Calculation

P = E/length(x);

% Display Results

disp(['Energy of the signal = ', num2str(E)]);

disp(['Power of the signal = ', num2str(P)]);

% Plot Signal

figure;

stem(n,x,'filled');

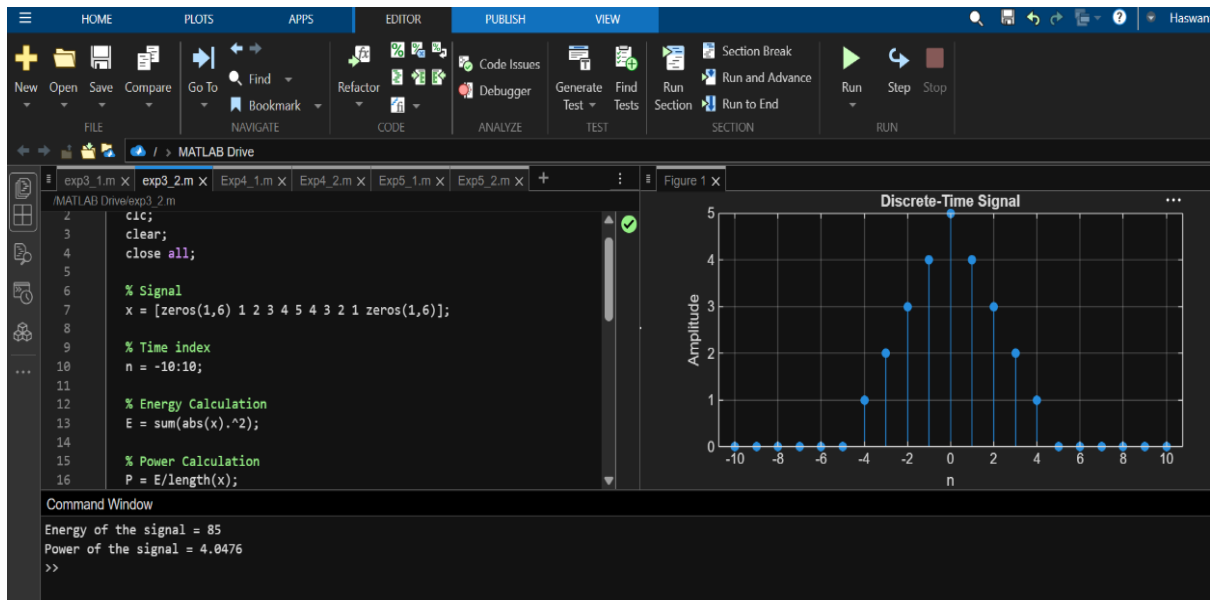
grid on;

```
title('Discrete-Time Signal');
```

```
xlabel('n');
```

```
ylabel('Amplitude');
```

Output Generated Graph:



Result:

The energy and average power of the given discrete-time signal were calculated successfully using MATLAB. The signal energy was found to be **85**, and the average power was **4.0476**.